

Predicting Weeks on Billboard for Rolling Stone's Top 500 Album

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Table of Contents

Summary	1
Introduction	2
Methodolgy	3
Importing the data	3
Preliminary Analysis	4
Loading The Necessary Libraries	4
Wrangling	4
Data Splitting	4
Exploratory Analysis	4
Predictor Selection and Model Execution	5
Discussion	6
References	7

Summary

Here, we attempt to build a regression model using the k-nearest neighbors (kNN) algorithm to predict the number of weeks an album spends on the Billboard chart `weeks_on_billboard` based on characteristics such as `release_year`, `peak_billboard_position`, and `spotify_popularity`. This model could help music industry analysts, artists, and record labels understand the factors that contribute to an album's longevity on the chart and make informed decisions about marketing and production strategies.

Our current model has a prediction error, as measured by the Root Mean Squared Error (RMSE), of approximately 72.16 weeks. This model has room for improvement, given that the `weeks_on_billboard` values has an Interquartile range of about 37 weeks.

Introduction

The data used to build this model comes from Rolling Stone’s rankings dataset, which includes information about album releases, chart performance, and artist details. The dataset was sourced from TidyTuesday, compiles information from Rolling Stone magazine’s renowned “The 500 Greatest Songs of All Time” lists. These lists were published in 2003, 2012, and 2020, each representing a snapshot of the most influential and celebrated songs according to music critics, artists, and industry experts at the time.

Variable	Class	Description
<code>sort_name</code>	character	Name used for sorting
<code>clean_name</code>	character	Clean name
<code>album</code>	character	Album name
<code>rank_2003</code>	double	Rank in 2003. NA if album not released yet or not in top 500
<code>rank_2012</code>	double	Rank in 2012. NA if album not released yet or not in top 500
<code>rank_2020</code>	double	Rank in 2020. NA if not in top 500
<code>differential</code>	double	2020-2003 Differential. Negative value if it went down in the chart. Positive value if it went up.
<code>release_year</code>	double	Release Year
<code>genre</code>	character	Album Genre
<code>type</code>	character	Album Type
<code>weeks_on_billboard</code>	double	Weeks on Billboard
<code>peak_billboard_position</code>	double	Peak Billboard Position
<code>spotify_popularity</code>	double	Spotify Popularity. NA if not on Spotify
<code>spotify_url</code>	character	Spotify URL. NA if not on Spotify
<code>artist_member_count</code>	double	Number of artists in the group
<code>artist_gender</code>	character	Gender of the artist(s). Male/Female if it’s a mixed-gender group
<code>artist_birth_year_sum</code>	double	Sum of the artists’ birth years
<code>debut_album_release_year</code>	double	Debut Album Release Year
<code>ave_age_at_top_500</code>	double	Average age at top 500 Album
<code>years_between</code>	double	Years Between Debut and Top 500 Album

Variable	Class	Description
album_id	character	Album ID. NOS at the beginning of the ID if not on Spotify

Part of the dataset shows the album rankings on the Billboard 200 Chart. The latter is a widely recognized measure of an album’s commercial success. Understanding the factors that contribute to an album’s longevity on the chart can provide valuable insights for artists, record labels, and music industry analysts. For example:

Artists: Can use this information to plan album releases and marketing strategies.

Record Labels: Can identify trends and allocate resources effectively.

Analysts: Can predict the potential success of new albums based on historical data.

In this project, we aim to answer the following question: Can we use characteristics of an album (e.g., release year, peak chart position, Spotify popularity) to predict how long it will stay on the Billboard chart?

To answer this question, we used a k-nearest neighbors (kNN) regression model and data from Rolling Stone’s rankings. We found that our model has a prediction error of approximately 72.16 weeks, which is reasonable given the variability in the data. However, there is room for improvement, as the model explains 18.67% of the variability in weeks_on_billboard.

Methodology

Importing the data

We imported the data set from TidyTuesday’s Github into the data directory. Using domain expertise and exploratory analysis, we chose to focus on the following predictor variables:

Table 1. Predictor variables used for analysis, chosen for their utility by domain expertise.

Predictor	Description
release_year	The year the album was released.
peak_billboard_position	The highest position the album reached on the Billboard chart.
spotify_popularity	A measure of the album’s popularity on Spotify (0–100 scale).
artist_gender	The artist’s gender.

Preliminary Analysis

Loading The Necessary Libraries

The necessary packages and libraries from the R programming language used in this analysis include `tidyr`, `dplyr`, `readr`, `ggplot2`, `ggpubr` and `caret`.

Wrangling

The data set used in this project was cleaned and prepared for analysis through several steps. First, numerical columns which had a large number of missing values (25% of total) and were poorly correlated as found in the EDA later were removed. Categorical variables like `sort_name`, `clean_name`, `album(name)`, `spotify_url`, etc. were removed, as they were not relevant to the analysis, and it would be difficult to use these variables as they are close to unique identifiers. Next, missing values were handled: the `genre` and `type` columns were imputed using the most frequent strategy, and the `weeks_on_billboard` column was imputed using the mean strategy, as removing these would have significantly skewed the analysis. Rows with missing values in other columns were dropped to ensure a complete dataset. Finally, all numeric predictors were scaled to ensure equal weighting in the kNN algorithm. These steps resulted in a clean, analysis-ready dataset.

Data Splitting

The data was partitioned into training (80%) and testing (20%) sets. The relationship between `weeks_on_billboard` and the predictor variables was visualized using scatter plots and boxplots to assess their utility for predictive modeling. Correlation was also used to assess the relationship between the response and the predictors.

Exploratory Analysis

In the exploratory data analysis (EDA), we first examined relationships between numeric variables and `weeks_on_billboard` using scatter plots with linear trendlines. This helped us visualize potential linear associations and identify predictors that might be strongly related to the target variable. Correlation values were also computed to quantify the strength of these relationships, highlighting which variables have positive or negative impacts on Billboard longevity.

Furthermore, we explored categorical variables — `genre`, `artist_gender`, and `type` — through boxplots to assess how `weeks_on_billboard` varies across different categories. These visualizations provided insights into how factors like album type and artist gender may influence chart

success, and revealed potential group-level differences that could be important for modeling. Overall, the EDA gave a clearer picture of key patterns and relationships in the data.

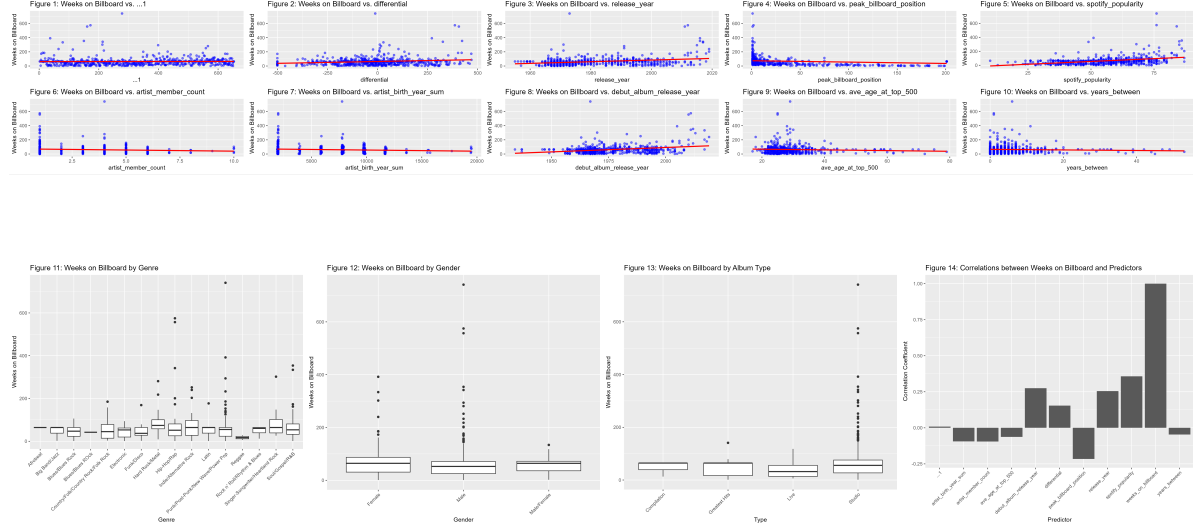


Figure 1: Scatterplots and boxplots for EDA

Predictor Selection and Model Execution

For predictor selection, we focused on identifying variables that had meaningful relationships with `weeks_on_billboard` and were relevant to understanding album success. Variables with an absolute correlation greater than 0.2 were prioritized, as these are more likely to contribute valuable information to the model. Based on this criterion, we selected `release_year`, `peak_billboard_position`, `spotify_popularity`, and `debut_album_release_year` as predictors. Variables with weak correlations, such as `differential`, `artist_member_count`, and `artist_birth_year_sum`, were excluded to prevent unnecessary complexity and reduce the risk of overfitting.

After selecting predictors, we cleaned and scaled the datasets using min-max normalization to ensure all numeric variables were on a comparable scale. We also handled mismatches in categorical levels between training and testing datasets by filtering out unseen categories

from the test set. A k-Nearest Neighbors (kNN) model was then trained using 5-fold cross-validation to tune the optimal value of k, evaluating performance with RMSE. The final model was assessed on the test set, where both RMSE and R-squared were calculated to measure prediction accuracy and explained variance. This process ensured a robust approach to selecting meaningful predictors and building a well-tuned predictive model.

kNN Model Tuning Results with Best k: 15
RMSE on Testing Set: 76.1993223621335 R-squared on Testing Set: 0.104860713627869

	k	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
1	1	76.2508498822922	0.145351745170446	41.0691251376828	25.7255246797427	0.0472367658255164	6.95054104310754
2	2	67.1646552856213	0.16113196810008	38.883825053152	19.5081798038812	0.0627370540057434	6.89439762122551
3	3	63.7267496273974	0.147302169196504	37.2186361021457	15.9621002020907	0.0664330661655985	5.09085169835028
4	4	63.1906976850103	0.132132897064161	36.8156390820718	16.9926206540763	0.0606895741730073	5.13726607944691
5	5	62.7837973856256	0.128302012129075	36.4862573644304	17.1159758029435	0.0667873460961372	4.7195432072484
6	6	62.2661742447849	0.122597791169274	35.9749770094069	17.1245198385585	0.0689977415818116	3.87677627680767
7	7	62.0701453108913	0.122236072132601	35.9136569245773	17.5131933044884	0.0701152518406793	3.96477498843331
8	8	61.7564311533424	0.113772246702669	35.8664923898217	18.4023268983942	0.0582279553914515	3.87073722797889
9	9	61.6306925350631	0.111819030772595	35.6110140849826	18.9470378594215	0.0553188708994166	3.8552191931637
10	10	61.8006479772288	0.102863578192881	35.6554137127441	19.3162379923189	0.0502725595670679	3.90578201040938
11	11	61.2524383309753	0.111421079596063	35.4405502592784	19.6874892934911	0.0379273431797488	4.10639977128198
12	12	61.100905994419	0.117429557376683	35.5234226190476	19.3489356303135	0.0415286945805018	3.98973418750316
13	13	61.2663972078179	0.112244037765863	35.6545361969321	19.3498264861408	0.0415024230648373	3.859808717128
14	14	61.3816265689879	0.10842431130453	35.8747266286827	19.6396277412127	0.0379193454372695	4.11655809904568
15	15	61.0915905696762	0.118970863976962	35.9056557654418	19.9169274143422	0.0341464748369059	4.04563223779443
16	16	61.2432432441006	0.113617132174789	36.127018746007	19.8842091703725	0.0419906288319471	4.14153262181255
17	17	61.5723381982664	0.10349237458889	36.3266508694506	19.9816435598705	0.0390670327881557	3.9582513212495
18	18	61.7934940046941	0.100204787284876	36.584641185962	20.0232439578049	0.0445273917821646	4.08575124209851
19	19	61.7607595848347	0.101116726334833	36.6102382953585	20.0627708565836	0.0497235959355512	4.17262999431106
20	20	61.5649574111007	0.104580167546436	36.4294146632519	20.1742822635018	0.0458376519681343	4.20875605343979

Figure 2: knn-tuning results

Discussion

Our interpretation of the data: The correlations between the predictors and the weeks on the billboard were generally weak. The variable that correlated with the weeks on the billboard most was the spotify popularity, which is expected since both are direct measurements of the popularity of the album. The correlation was 0.35723461. The variable that correlated with the weeks on the billboard the least was the years between the release date of the album and the first album the artist released. This was a bit surprising since people might expect more experienced artists to make more popular works. The correlaton was -0.04631483.

Implications of the data: The data could be used to predict the peak productivity of artists and determine which artists show the most promise in regard to their future career opportunities. Our data seems to imply that artists peak early in their careers. This may imply that the

conditions required to produce popular music are mainly circumstantial and are difficult to recreate. This may raise future questions such as whether hiring based on experience is a good practice in some industries.

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